Honors Focus Area & Capstone Reflection (non-EGL) – Tianyi Chen

1. Rank the capstone types in order of your current interest:

Research	_____1____
Design	_____3____
Entrepreneurship	_____2____
Global Business/Operations	_____5____
Public Service	_____4____

2. How did you decide which choice to rank as #1 above?

I love pursuing research and I ranked it as my #1 choice because research in so interesting by the fact that it is so niche, most of the time, the thing you are researching no one has ever done before, and you are literally one of the first person to touch that particular subject, or research area, and that has always sounded really awesome and exciting for me, just to work on something that no one has ever done before.

3. Brainstorm 2-3 ideas for your Honors Capstone project. What type of project would be most interesting to you? What sort of project experience would be beneficial to you in the future? These can be structured, semi-structured, or independently created project ideas.

For my Honors Capstone project, would be interested in doing a capstone project on one of the research projects I am currently doing, I am currently working in a robotics lab and an automotive lab, in which both closely relates to mechanical engineering.

One project I am working on the making a circuit board that functions as a sensor for testing flame propagation inside the piston combustion chamber. Another project I am working on is designing a fuel tank for a combustion-driven micro insect scale micro-actuator. I would be interested in including one of those projects as my Capstone project for Honors.

The experience that would benefit me in the future would be improved PCB designing skills, and also combustion testing, along with CAD skills.

4. After reviewing the Focus Area FAQs and Selecting a Theme and Finding Courses Canvas pages, list three specific focus area themes you would be interested in pursuing. Make sure at least one of

these themes is a custom theme.

After reviewing both documents, the pathway I would like to take it:

Product Design, Manufacturing, and Materials in engineering.
Healthcare Innovation and Biomedical Systems in medical sciences.
Custom theme: AI integration in robotics, and control systems.

5. In SMART goal format, list three specific things you will do to explore, identify, or finalize your focus area or capstone project ideas.

   By November 15, 2026, I will meet with my research advisors from both the robotics lab and automotive lab to discuss the possibility of using my current research projects for my Honors Capstone.

   By December 1, 2026, I will further investigate what specific area of research I would like to include in my Capstone project/useful to include, from both projects I am currently working on. I will review at least five relevant research papers and compare them to identify potential impact of each project.

   By December 15, 2026, I will select one of the two research projects as my Honors Capstone focus based on my interests and the feedback I get from my advisor, then, I will plan a rough draft of what my Capstone poster will look like in terms of what sections it should have.